# Personalised Practice: Resultant Forces

Topic: Forces and Motion

Original examination-style practice material

Total: 11 marks

### 1 12 N forwards

Resultant force = 18 - 6 = 12 N.

Marks: subtracts opposing force · 12 N · forwards [3]

### 2 (a) 1800 N forwards (b) 2.0 m/s²

Resultant force = 2400 - 600 = 1800 N.

$a = F/m = 1800/900 = 2.0\ \text{m/s}^2$.

Marks: subtracts forces · resultant · selects F = ma · substitution · answer with unit [5]

Accept: $2\ \text{m/s}^2$

### 3 Weight equals air resistance, so the resultant force and acceleration are zero; the parachutist continues at constant velocity.

Marks: forces balanced · zero acceleration · constant velocity [3]